````
---
title: Cool Mix Flows
toc: false
---

<link rel="stylesheet" href="./styles/App.css">

```js
import {OVERRIDE_YEAR, RIVER_MILE} from "./config.js";
```

```js
const toF = c => c * 9 / 5 + 32;
```

```js
const raw = await FileAttachment("./data/RM15_water_temp.csv").text();
const parsed = d3.csvParse(raw, d => ({
  date: d3.timeParse("%Y-%m-%d")(d.date),
  tmp: d.tmp === "" ? null : toF(+d.tmp),
  cfs: d.cfs === "" ? null : +d.cfs
}));
```

```js
// Hand-maintained lookup (data/cool_mix_flows.csv) ? on/off-ramp dates
// are operational decisions, not measured data, so they live separately
// from the R data loader. Blank onRamp/offRamp = Cool Mix did not run.
const coolMixRaw = await FileAttachment("./data/cool_mix_flows.csv").text();

// Date format flexibility
function parseFlexDate(s) {
  if (!s) return null;
  return d3.timeParse("%Y-%m-%d")(s) ?? d3.timeParse("%m/%d/%Y")(s);
}

const coolMix = d3.csvParse(coolMixRaw, d => ({
  year: +d.year,
  onRamp: parseFlexDate(d.onRamp),
  offRamp: parseFlexDate(d.offRamp),
  targetRM: d.targetRM ? +d.targetRM : null,
  costUSD: d.costUSD ? +d.costUSD : null,
}));
```

```js
// Only the years Cool Mix actually ran
const coolMixYears = coolMix
  .filter(d => d.onRamp && d.offRamp)
  .sort((a, b) => a.year - b.year);
```

```js
const threshold = toF(15.5);
const doyOf = d => d3.utcDay.count(d3.utcYear(d), d);
const monthNames = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"];
const monthStarts = [0, 31, 59, 90, 120, 151, 181, 212, 243, 273, 304, 334];
```

```js
// Fixed color per Cool Mix year
const palette = ["#3987e5", "#e34948", "#1baf7a", "#9085e9"];
const yearColors = Object.fromEntries(coolMixYears.map((d, i) => [d.year, palette[i % palette.length]]));
```

```js
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````
const tempByYear = coolMixYears.map(({year}) => ({
  year,
  rows: parsed
    .filter(d => d.date && d.date.getFullYear() === year && d.tmp !== null)
    .map(d => ({...d, doy: doyOf(d.date)}))
}));

const bandData = coolMixYears.map(d => ({
  year: d.year,
  x1: doyOf(d.onRamp),
  x2: doyOf(d.offRamp),
}));
```

```js
const currentYear = OVERRIDE_YEAR ?? d3.max(parsed.filter(d => d.date).map(d => d.date.getFullYear()));
const logo = await FileAttachment("assets/LOGO.png").url();

display(htl.html`
<header class="site-header">
  <img src="${logo}" alt="Grand Canyon Trust" class="site-logo">
  <nav class="site-nav">
    <a href="./">Overview</a>
    <a href="./season">${currentYear} Season</a>
    <a href="./cool-mix" aria-current="page">Cool Mix Flows</a>
  </nav>
</header>
`);
```

```js
display(htl.html`<div class="hero">
  <div class="hero-bg"></div>
  <p class="hero-eyebrow">Colorado River Mile ${RIVER_MILE} · Grand Canyon National Park</p>
  <h1 class="hero-title">Cool-Mix Flows</h1>
  <p class="hero-sub">A strategy to protect native fish in the Grand Canyon</p>
</div>`);
```

```js
const chartPlot = resize((width) => Plot.plot({
  width,
  height: 480,
  marginLeft: 60,
  marginBottom: 45,
  style: {
    fontFamily: "IBM Plex Mono, monospace",
    fontSize: "16px",
    background: "transparent",
  },
  x: {
    domain: [0, 365],
    ticks: monthStarts,
    tickFormat: doy => {
      const i = monthStarts.indexOf(doy);
      return i >= 0 ? monthNames[i] : null;
    },
    label: null,
  },
  y: {
    label: "Water temperature (°F)",
    domain: [toF(7), toF(22)],
  },
  marks: [
    // Cool Mix window, one band per year, colored to match that year's line
    ...bandData.map(d => Plot.rectY([d], {
````

````
      x1: "x1", x2: "x2", y1: toF(7), y2: toF(22),
      fill: yearColors[d.year], fillOpacity: 0.10,
    })),
    Plot.ruleY([threshold], {
      stroke: "#B03823", strokeDasharray: "4 2", strokeWidth: 1.2,
    }),
    Plot.text([{}], {
      x: 364, y: threshold, text: () => `${threshold.toFixed(1)}°F threshold`,
      dy: -8, textAnchor: "end", fontSize: 13, fontWeight: 500, fill: "#B03823",
    }),
    ...tempByYear.map(({year, rows}) => Plot.line(rows, {
      x: "doy", y: "tmp",
      stroke: yearColors[year], strokeWidth: 2.5,
    })),
    Plot.tip(
      tempByYear.flatMap(({year, rows}) => rows.map(d => ({...d, year}))),
      Plot.pointerX({
        x: "doy", y: "tmp",
        title: d => `${d.year} ? ${d.date.toLocaleDateString("en-US", {month: "short", day: '
      })
    ),
  ],
}));
display(htl.html`<div class="chart-card">${chartPlot}</div>`);
```

```js
const legendEl = htl.html`<div class="legend">
  <div class="legend-items">
    ${coolMixYears.map(d => htl.html`<div class="legend-item">
      <svg width="32" height="12">
        <line x1="0" y1="6" x2="32" y2="6" stroke="${yearColors[d.year]}" stroke-wi
dth="2.5" />
      </svg>
      <span>${d.year} (Cool-Mix active ${d.onRamp.toLocaleDateString("en-US", {mont
h: "short", day: "numeric"})}?${d.offRamp.toLocaleDateString("en-US", {month: "short", day: '
an>
    </div>`)}
    <div class="legend-item">
      <svg width="32" height="12">
        <line x1="0" y1="6" x2="32" y2="6" stroke="#B03823" stroke-width="1.5" stro
ke-dasharray="4,2"/>
      </svg>
      <span>${threshold.toFixed(1)}°F Threshold</span>
    </div>
  </div>
</div>`;
display(legendEl);
```

<p class="last-updated">
Cool-mix flows began in 2024 as a tool to reduce water temperatures in the Colorado
 River through the Grand Canyon. To cool the river below Glen Canyon Dam, managers
release colder water through the lowest outlets deep within Lake Powell. Cold tempe
ratures reduce the ability of nonnative fish, like smallmouth bass, to reproduce an
d establish populations. Smallmouth bass prey on and compete with native fish, like
 humpback chub, and their expansion could threaten native fish in the Grand Canyon.
</p>

<p class="last-updated">
In 2024 and 2025, managers conducted cool-mix flow experiments to lower water tempe
ratures and prevent smallmouth bass from expanding into Grand Canyon National Park.
 The graph shows how water temperatures changed during the cool-mix flow periods, m
arked by the shaded bands. In both years, cool-mix flows brought temperatures down
during periods when they would have exceeded the 59.5 °F (15.5 °C) threshold. No sm
allmouth bass reproduction was detected during either year.
</p>
````
````

````
<p class="last-updated">
In 2026, water managers decided not to implement cool-mix flows.
</p>

```js
// ?? Compact reference table ??????????????????????????????????????????
display(htl.html`<div class="section-card section-card--table">
  <div class="section-card-header">
    <h2 class="section-card-title">Cool-Mix Reference</h2>
  </div>
  <div class="recent-table-scroll">
  <table class="recent-table">
    <thead>
      <tr><th>Year</th><th>On-Ramp</th><th>Off-Ramp</th><th>Duration</th><th>Target RM</th><th>Cost</th></tr>
    </thead>
    <tbody>
      ${coolMixYears.slice().reverse().map(d => htl.html`<tr>
        <td>${d.year}</td>
        <td>${d.onRamp.toLocaleDateString("en-US", {month: "short", day: "numeric"})}</td>
        <td>${d.offRamp.toLocaleDateString("en-US", {month: "short", day: "numeric"})}</td>
        <td>${d3.timeDay.count(d.onRamp, d.offRamp)} days</td>
        <td>${d.targetRM}</td>
        <td>${d.costUSD ? d3.format("$,.2s")(d.costUSD) : "?"}</td>
      </tr>`)}
    </tbody>
  </table>
  </div>
</div>`);
```

```js
display(htl.html`
  <footer class="site-footer">
    <p class="site-footer-citation">
      Observations from
        <a href="https://waterdata.usgs.gov/monitoring-location/USGS-09380000" target="_blank" rel="noopener noreferrer">USGS Gage 09380000</a>
        · Temperature estimates based on
        <a href="https://doi.org/10.1002/eap.2279" target="_blank" rel="noopener noreferrer">Dibble et al. (2020)</a>
        thermal model · PLACEHOLDER DATA CITATION FOR COOLMIX FLOWS.
    </p>
    <p class="site-footer-credit">
      Graphs may be used for non-commercial purposes provided that they are not altered or edited and are appropriately credited to the Grand Canyon Trust.
    </p>
  </footer>
`);
```
````